Halil Çağrı Bilgi

PhD Candidate in Graph Machine Learning — Delft University of Technology

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SUMMARY

PhD candidate working on graph neural networks across expressivity, adversarial robustness, privacy-preserving inference, and explainability, applied to financial crime detection on edge-attributed multigraphs. Published at top-tier machine learning venues, with open-source implementations. Over four years of research and engineering experience across academia and industry.

EDUCATION

PhD, Computer Science Jun 2024 – Present

Delft University of Technology, Delft, Netherlands

- Data-Intensive Systems group. Daily supervisor: Dr. Kubilay Atasu. Promotor: Prof. Dr. Geert-Jan Houben.
- Thesis topic: trustworthy graph neural networks for anti-money laundering on multigraph transaction networks.
- Expected completion: 2028.

MSc, Electrical and Electronics Engineering Sep 2021 – Apr 2024

Middle East Technical University (METU), Ankara, Turkey

- Specialized in computer vision and signal processing.
- Thesis on graph neural networks for multi-object tracking, supervised by Prof. Dr. Aydın Alatan.

BSc, Electrical and Electronics Engineering Feb 2017 – Jul 2021

Middle East Technical University (METU), Ankara, Turkey

- Senior-year specialization in signal processing and machine learning.

RESEARCH EXPERIENCE

Doctoral Researcher Jun 2024 – Present

Delft University of Technology, Delft, Netherlands

Research on graph neural networks for financial crime detection, spanning expressivity of message passing on multigraphs, adversarial robustness, and privacy-preserving inference under homomorphic encryption. See publications below. Current work addresses explainability of GNN decisions for financial crime analysts, including the use of large language models.

Research Assistant Nov 2022 – May 2024

Middle East Technical University (METU), Ankara, Turkey

- Formulated multi-object tracking as link prediction on a graph to improve robustness under occlusion. Published at IEEE ICIP 2024.

Research Engineer Sep 2021 – Oct 2023

METU Center for Image Analysis, Ankara, Turkey

- Developed a graph-based deep learning method to merge fragmented target trajectories in multi-object tracking, and applied optimal transport theory to tracklet merging.
- Built an end-to-end pipeline for orbit determination, propagation, and refinement to estimate positions and orbits of space objects from imagery, for an international project funded by Telespazio.

PUBLICATIONS

Towards Adversarially Robust Multigraph Neural Networks for Financial Crime Analysis
Dennis Heijmans, H. Çağrı Bilgi, Rui Wang, Kubilay Atasu. *Learning on Graphs Conference (LoG)*, 2026.

Random Probing for Structural Self-Interactions in Graph Neural Networks
Saku Peltonen, H. Çağrı Bilgi, Kubilay Atasu. *Learning on Graphs Conference (LoG)*, 2026.

Leveraging Time-Series Foundation Models for Financial Crime Detection
Danae Natalie Savvidi, H. Çağrı Bilgi, Kubilay Atasu, Arie van Deursen. *7th International Conference on AI in Finance (ICIAF)*, 2026.

Privacy-Preserving GNN Inference on Financial Transaction Graphs via Homomorphic Encryption
H. Çağrı Bilgi, Kemal Mutluergil, Ceren Yıldırım, Kubilay Atasu, Erkay Savaş. *Workshop on Financial AI Security, Privacy, and Safety, International Conference on AI in Finance (ICIAF)*, 2026.

MEGA: Message Passing Neural Networks for Multigraphs with Edge Attributes
H. Çağrı Bilgi, Kubilay Atasu. *Transactions on Machine Learning Research (TMLR)*, 2026.

Bi-Directional Tracklet Embedding for Multi-Object Tracking
H. Çağrı Bilgi, A. Aydın Alatan. *IEEE International Conference on Image Processing (ICIP)*, 2024.

SKILLS

Expertise: Graph neural networks, expressivity of message passing, adversarial robustness, privacy-preserving machine learning, explainability, computer vision

Machine learning: LLM fine-tuning, distributed and multi-GPU training, large-scale graph processing

Tools: Python, PyTorch, PyTorch Geometric, Weights & Biases, HPC clusters (SLURM), Docker, Git, LaTeX

Languages: English (fluent), Turkish (native)